

“Smart Attendance Management System”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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Declaration

The Project Report entitled “**Face Attendance Management System**” is a record of bonafide work of **V.Shiva Venkata Sai – 2520090092, R.Pranav – 2520090109, V.Sathvik - 2520090092** submitted in partial fulfillment of the requirements of the course **DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)**, Section **10**, Team **27**, during **2026–2027, Trimester 4**, at **KLH University**. The results embodied in this report have not been copied from any other student, department, or institution.

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CERTIFICATE

This is to certify that the project report entitled “**Face Attendance Management System**” is a bonafide work done and submitted by **V. Shiva Venkata Sai – 2520090092, R. Pranav – 2520090109, V. Sathvik – 2520090097** in partial fulfillment of the requirements of the course **DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)**, Section 10, Team 27, at **KLH University**, during **2026–2027, Trimester 4**.

Signature of the Guide

Signature of the Course Coordinator

Signature of the HOD

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my Course Coordinator **Dr** , and my guide **Divya Priya Degala**, Department of Computer Science and Information Technology which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. Kaja Shareef**, Head of the **Department for Computer Science and Information Technology** for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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ABSTRACT

The **Face Attendance Management System** is a web-based application designed to automate and simplify the process of recording and managing student attendance using **facial recognition technology**. Traditional attendance methods, such as manual roll calls and paper-based records, are time-consuming, prone to errors, and difficult to maintain efficiently. This project provides a digital solution that enables attendance to be recorded quickly and accurately by identifying students through their facial features.

The system allows authorized users to log in and access the attendance dashboard, where student information and attendance records can be managed. The facial recognition module captures and identifies registered students, after which their attendance is automatically recorded in the database. The system also provides information such as the **total number of students, the number of students present, and the names of students who are present**. This makes it easier for faculty to monitor attendance in real time.

The application uses a **database management system** to store student details and attendance records securely and systematically. A web-based frontend provides an easy-to-use interface, while the backend handles authentication, database operations, and attendance processing. The system reduces manual effort, minimizes attendance errors, and provides faster access to attendance information.

Overall, the **Face Attendance Management System** demonstrates how facial recognition, web technologies, and database management can be integrated to create an efficient, user-friendly, and automated attendance solution for educational institutions.

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CHAPTER -1

1.1 Background of the Project

Educational institutions need to maintain accurate attendance records for students on a regular basis. Traditionally, attendance is recorded manually through roll calls, attendance sheets, or spreadsheets. These methods require significant time and effort from faculty members and may lead to errors such as incorrect marking, duplicate entries, or difficulty in maintaining records.

The **Face Attendance Management System** was developed to automate this process using facial recognition technology. Instead of manually calling student names, the system identifies registered students through their facial features and records their attendance digitally. The attendance information is stored in a database, allowing authorized users to access and monitor attendance records efficiently.

The system also provides an attendance dashboard that displays the **total number of students, the number of students present, and the names of students who are present**. This provides a clear and centralized view of attendance and reduces the effort required for manual attendance management.

1.2 Problem Statement

In a traditional classroom environment, faculty members need to manually identify students and record their attendance. This process can be time-consuming, particularly when the number of students is large. Manual attendance can also be affected by human errors, duplicate entries, and difficulties in maintaining and retrieving historical attendance records.

The system therefore needs to provide an automated method for identifying registered students, recording their attendance, preventing repeated attendance marking, and displaying updated attendance information. The system should also ensure that only authorized users can access and manage attendance-related information.

1.3 Scope of the Project

The **Face Attendance Management System** covers the process of identifying students through facial recognition and maintaining their attendance records in a centralized database. The system includes student registration, user authentication, facial recognition, attendance recording, and attendance monitoring.

The system provides a web-based interface through which authorized users can log in and access the attendance dashboard. The dashboard displays the total number of students, the number of students present, and the names of students who have been marked present. Attendance information is stored systematically in the database for future reference.

The project focuses on automated attendance management using facial recognition. Advanced features such as mobile applications, cloud deployment, automated notifications, and advanced attendance analytics may be considered outside the current scope and can be included as future enhancements.

1.4 Objectives

- **Automate student attendance** using facial recognition technology.
- **Reduce the time and effort** required for manual attendance marking.

- **Minimize human errors** associated with traditional attendance methods.
- **Identify registered students** accurately using their facial information.
- **Store attendance records** in a centralized database for easy access and management.
- **Display the total number of students** registered in the system.
- **Display the number of students present** for the current attendance session.
- **Display the names of students who are present** on the attendance dashboard.
- **Prevent duplicate attendance entries** for the same student during an attendance session.
- **Provide authenticated access** so that attendance information can only be accessed by authorized users.

1.5 Importance of the Application

The **Face Attendance Management System** helps educational institutions reduce the manual effort involved in attendance management. By using facial recognition, students can be identified automatically, allowing attendance to be recorded more quickly than traditional roll-call methods.

The centralized database makes attendance records easier to store, retrieve, and maintain. Faculty members can use the dashboard to immediately view the total number of students and identify which students are present. This improves the visibility of attendance information and reduces dependence on paper-based records or manually maintained spreadsheets.

The application also provides a foundation for integrating additional attendance-related features in the future, such as attendance percentage calculations, reports, notifications, and automated check-in mechanisms.

1.6 Target Users / Audience

The system primarily serves **faculty members, administrators, and students**.

Students are registered in the system and identified through facial recognition when attendance is recorded. Their attendance information is stored against their respective student records.

Faculty members can use the system to conduct attendance sessions, view the attendance dashboard, and monitor which students have been marked present.

Administrators can manage student information, authorized users, and attendance-related data, depending on the access permissions implemented in the system.

The system is therefore intended for **educational institutions such as colleges and universities** that want to automate and simplify their student attendance management process.

CHAPTER 2: SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The Face Attendance Management System has modest hardware requirements, since the application is designed to handle a relatively small number of students and attendance records. A development or demonstration computer with a dual-core processor, 4 GB of RAM, and approximately 500 MB of free disk space is sufficient to run the frontend, backend, database, and required project files on a single machine. A webcam or built-in camera is also required for capturing the student's face during the attendance process.

2.2 Software Requirements

- Node.js 18 or later, with npm
- MySQL 8.0 or later
- A modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox
- Git, for version control
- Visual Studio Code, for development
- MySQL Workbench, for creating, managing, and inspecting the database

2.3 Development Tools and Frameworks

The Face Attendance Management System was developed as a web-based application using a frontend, backend, and relational database. Node.js was used to run the backend services, while the web frontend provides the interface for user authentication, face-based attendance, and attendance monitoring. MySQL was used as the relational database for storing student information and attendance records. The MySQL database is accessed from the backend using a suitable MySQL driver. Visual Studio Code was used as the primary development environment, while MySQL Workbench was used for creating, inspecting, and verifying the database schema and stored attendance data. Git was used for source-code version control and project management.

CHAPTER 3: TECHNOLOGY STACK

3.1 Frontend

The frontend of the Face Attendance Management System provides the user interface through which users can log in, access the attendance system, and view attendance information. The interface is designed to provide a simple and responsive experience for students and administrators. The frontend communicates with the backend through HTTP requests and displays information such as the total number of registered students, the number of students present, and the names of students whose attendance has been recorded. CSS is used to provide the visual layout, styling, forms, buttons, cards, tables, and dashboard components of the application.

3.2 Backend

The backend of the Face Attendance Management System handles the application's server-side operations and acts as the communication layer between the frontend and the database. It manages user authentication, student information, attendance processing, and requests received from the frontend. The backend validates incoming requests and performs the required database operations before returning the appropriate response to the frontend. This separation between the frontend and backend keeps the application organized and allows database operations to be handled securely on the server side.

3.3 Database

MySQL is used as the relational database for storing the persistent information required by the Face Attendance Management System. The database stores student details, user authentication information, and attendance records. Relationships between the stored information are maintained using appropriate primary keys and foreign keys. Attendance data can therefore be retrieved efficiently to determine the students who are present and to maintain attendance records for future reference.

3.4 Version Control

Git is used throughout the development of the Face Attendance Management System for version control. The project source code can be maintained in a Git repository, allowing changes to the frontend, backend, and database-related files to be tracked during development. GitHub can be used to host the repository and provide a centralized location for maintaining and sharing the project source code among team members.

3.5 Authentication and Security Libraries

Authentication is implemented to restrict access to the attendance system to authorized users. Login credentials are verified by the backend before access to protected functionality is provided. Passwords are stored in the database in protected form rather than as plain-text values. Authentication also helps separate normal user functionality from administrative functionality where required. Server-side validation is used to ensure that unauthorized requests cannot directly access protected attendance or student information.

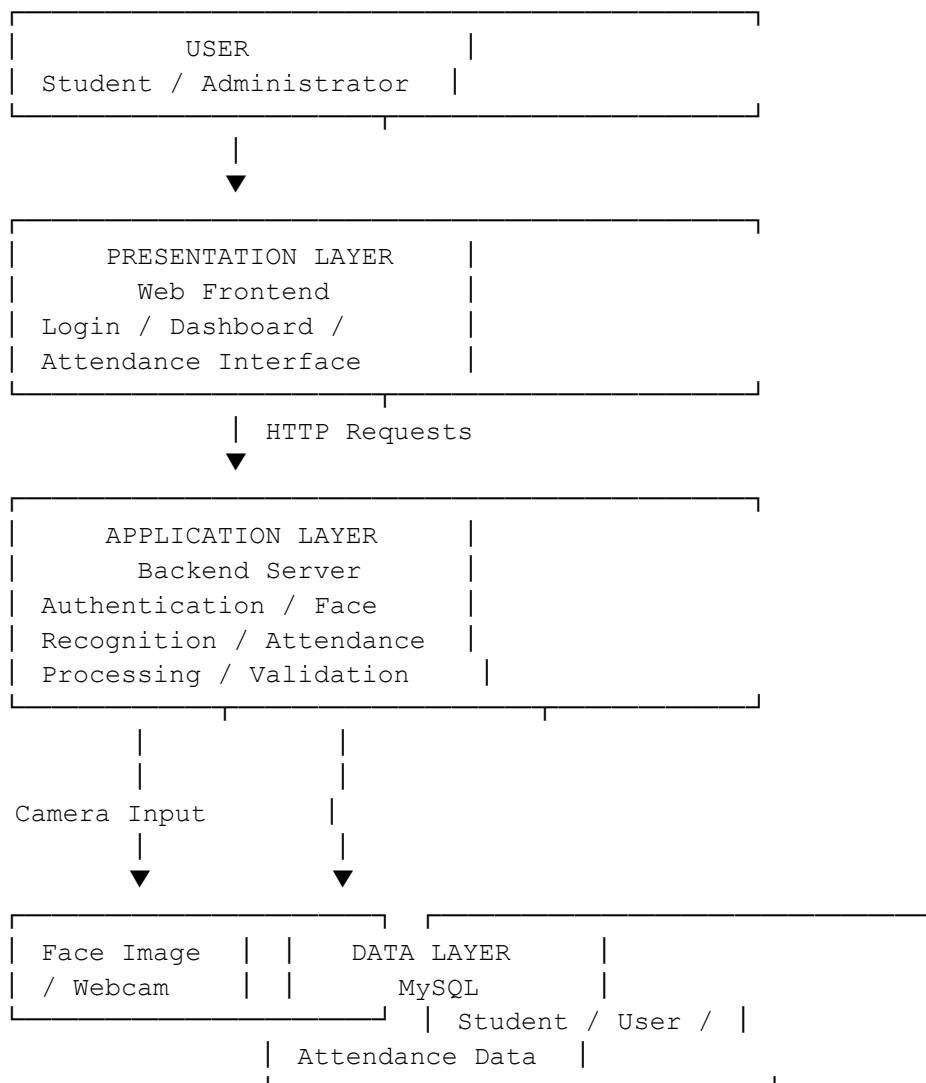
CHAPTER 4: SYSTEM ARCHITECTURE

4.1 High-Level Architecture Diagram

The Face Attendance Management System follows a layered architecture in which the user interface, application logic, and database are separated into different logical layers. The architecture allows users to interact with the system through the frontend, while the backend processes requests and communicates with the MySQL database.

The major components of the system are the **Presentation Layer**, **Application Layer**, and **Data Layer**. A camera is also used as an input component during the face-based attendance process. Figure 4.1 represents the overall flow of the Face Attendance Management System.

Figure 4.1: High-Level Architecture of Face Attendance Management System



4.2 Description of Each Layer

The **Presentation Layer** provides the web interface through which users interact with the Face Attendance Management System. It contains the login interface, attendance interface, and dashboard components. The frontend sends requests to the backend and displays the responses received from the server. It does not directly communicate with the MySQL database.

The **Application Layer** consists of the backend server and contains the main application logic. It receives requests from the frontend, validates user information, processes attendance-related operations, and communicates with the database. During the attendance process, the application also handles the face-related input required to identify a registered student and record their attendance.

The **Data Layer** consists of the MySQL database. It stores the persistent information required by the system, including user details, student information, and attendance records. The backend performs database operations such as inserting new attendance records and retrieving attendance information for display on the dashboard.

The **Camera Input** acts as an input source for the face attendance process. The camera captures the student's face, which is processed by the application's face-recognition functionality to identify the corresponding registered student.

4.3 Deployment Architecture

For development and demonstration purposes, the Face Attendance Management System can be operated on a single computer. The frontend runs through the web application, the backend server handles application requests, and MySQL runs as the database server. A webcam or built-in camera is used as the input device for capturing faces during attendance.

The system can initially be deployed in a local development environment for testing and demonstration. The architecture can later be extended to a hosted environment in which the frontend, backend, and database are deployed on suitable servers or cloud infrastructure.

CHAPTER 5: DESIGN

5.1 Database Design Principles

The database of the Face Attendance Management System is designed to store student and attendance information in an organized relational structure. Student information and attendance information are maintained separately so that the same student details do not need to be repeatedly stored for every attendance record.

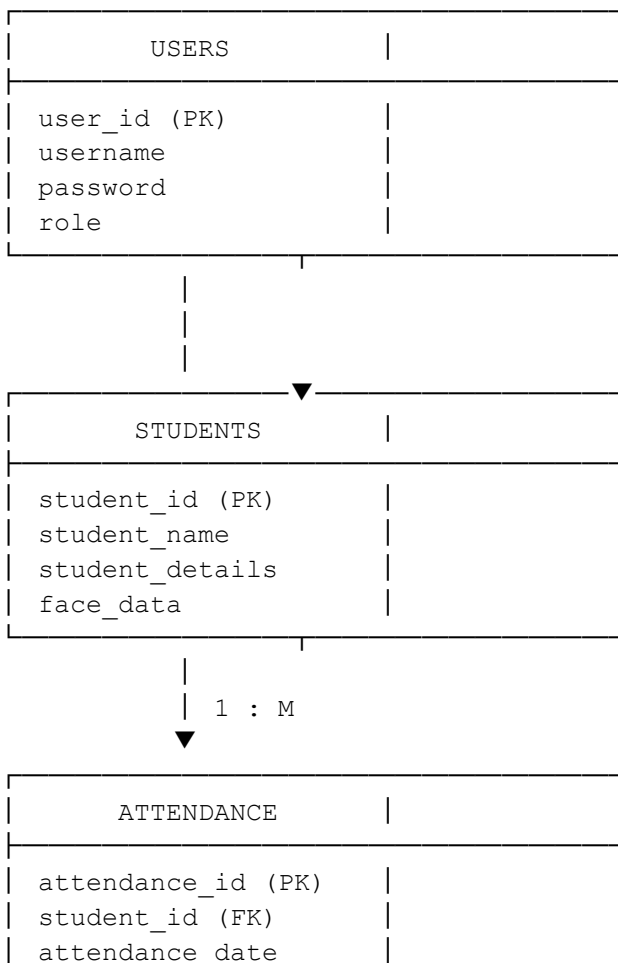
Each student is identified using a unique identifier, which can be referenced by attendance records. Relationships between tables are maintained using primary keys and foreign keys where applicable. Appropriate constraints are used to maintain data consistency and prevent invalid or duplicate records.

The database design also supports retrieving attendance information efficiently. This allows the application to calculate the total number of students, determine the number of students marked present, and retrieve the names of students who have attended.

5.2 ER Diagram / Database Schema

The database consists of related entities required for user authentication, student management, and attendance recording. The major entities of the system include **Users**, **Students**, and **Attendance**.

Figure 5.1: ER Diagram / Database Schema



	attendance_time	
	status	

The **USERS** entity stores the authentication information required to access the application. The **STUDENTS** entity stores information about students registered in the attendance system and the information required for face-based identification. The **ATTENDANCE** entity stores the attendance records generated when a student is successfully identified.

The relationship between Students and Attendance allows one student to have multiple attendance records across different dates or attendance sessions. The student identifier stored in the attendance table connects each attendance entry with the corresponding student.

5.3 Design Decisions and Assumptions

The system follows several design decisions to keep attendance information consistent and easy to manage:

- **Separate student and attendance information:** Student details are stored once, while individual attendance events are stored separately. This avoids repeatedly storing the same student information.
- **Unique student identification:** Each registered student is assigned a unique identifier so that the system can correctly associate a recognized face with the corresponding student record.
- **Attendance duplication control:** The system is designed to prevent the same student from being unnecessarily marked present multiple times for the same attendance session or date, depending on the attendance rules implemented by the application.
- **Authenticated access:** Login is required before users can access protected parts of the application. This prevents unauthorized users from directly accessing student and attendance information.
- **Dashboard-based monitoring:** Attendance information is retrieved from the database and summarized on the dashboard. The dashboard displays the total number of students, the number of students present, and the names of students marked present.

CHAPTER 6: IMPLEMENTATION

6.1 Module-wise Implementation

The Face Attendance Management System is organized into several functional modules that work together to automate the attendance process. The major modules include **Authentication**, **Student Management**, **Face Recognition**, **Attendance Management**, and **Dashboard**.

The **Authentication Module** manages user login and access control. It verifies the credentials submitted by the user before providing access to the protected application.

The **Student Management Module** handles student information stored in the database. It provides the information required to identify students and associate their face data with the corresponding student record.

The **Face Recognition Module** handles the face-based identification process. During attendance, the student's face is captured through the camera and processed by the face-recognition component. When a registered student is successfully identified, the corresponding student information is retrieved.

The **Attendance Management Module** records the attendance of the identified student in the database. It associates the attendance record with the appropriate student and stores relevant attendance information such as the date, time, and attendance status.

The **Dashboard Module** retrieves attendance information from the database and presents it in an understandable format. It displays the total number of students, the number of students currently marked present, and the names of students whose attendance has been recorded.

6.2 Front-end Logic

The frontend provides the screens through which users interact with the Face Attendance Management System. The login interface accepts the user's credentials and communicates with the backend for authentication. After successful authentication, the user can access the appropriate dashboard and attendance functionality.

The attendance interface provides the camera-based interaction required for face identification. Once a face is recognized, the frontend receives the result from the backend and displays the identified student's information and attendance status.

The dashboard retrieves information from the backend and dynamically displays attendance statistics. The total student count, present student count, and names of present students are obtained from the database rather than being manually entered into the interface.

6.3 API Integration

The frontend communicates with the backend through HTTP API requests. These requests are used for operations such as user login, retrieving student information, processing attendance, and obtaining dashboard statistics.

The backend receives each request, performs the required validation and processing, communicates with the MySQL database when necessary, and returns the result to the frontend. This approach ensures that database operations remain on the server side and that the frontend receives only the information required for the corresponding operation.

The attendance API is responsible for processing the identified student and creating the corresponding attendance record. Dashboard-related API requests retrieve the total number of students, the number of students marked present, and the names of students who are present.

6.4 Authentication and Authorization

Authentication is used to ensure that only registered users can access the protected functionality of the Face Attendance Management System. During login, the credentials submitted by the user are sent to the backend for verification.

If the credentials are valid, the backend provides the authentication response required by the frontend to maintain the user's authenticated session. Protected operations are verified by the backend before they are processed.

Authorization can also be used to distinguish between different types of users, such as administrators and normal users. Administrative functionality can be restricted to authorized accounts, while normal users can be given access only to the functions required for attendance-related activities.

This approach ensures that student information and attendance records are not directly accessible through unauthorized requests and that the application's access rules are enforced on the server side.

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CHAPTER 7: FEATURES

7.1 List of Main Features

The Face Attendance Management System provides the following major features:

Student/User-facing features:

- User registration and secure login
- Authentication-based access to the attendance system
- Face-based student identification using a camera
- Automatic attendance recording after successful face recognition
- Display of the identified student's name after recognition
- Prevention of unnecessary duplicate attendance entries for the same attendance session
- View attendance status after successful identification
- Access to attendance information through the web interface

Administrator/System-facing features:

- Register and maintain student information
- Store student information required for face-based identification
- Capture or process a student's face through the camera
- Automatically identify registered students during attendance
- Record the date and time of attendance
- Display the total number of registered students
- Display the total number of students currently marked present
- Display the names of students who are present
- Retrieve attendance records from the database
- Provide a centralized dashboard for monitoring attendance
- Maintain attendance information in the MySQL database for future reference

7.2 How Each Feature Works

User Authentication:

When a user enters their login credentials, the frontend sends the information to the backend for verification. The backend validates the credentials against the stored user information in the

database. If the credentials are valid, the user is allowed to access the protected functionality of the application. Invalid credentials are rejected and an appropriate error message is displayed.

Face-based Attendance:

The attendance process uses a camera to capture the student's face. The captured face is processed by the face-recognition component of the application and compared with the registered student information. When a matching student is identified successfully, the system retrieves the corresponding student record and records the student's attendance.

Student Name Display:

After successful identification, the system retrieves the student's name associated with the recognized student record. The name is displayed on the attendance interface so that the user can clearly verify which student has been identified.

Automatic Attendance Recording:

Once a registered student is successfully recognized, the backend creates an attendance record in the database. The record contains the information required to identify the student and the corresponding attendance date and time. This eliminates the need for manually entering the student's name during attendance.

Attendance Dashboard:

The dashboard retrieves student and attendance information from the database and presents it in a summarized form. The dashboard displays the total number of registered students, the number of students who are present, and the names of students whose attendance has already been recorded.

Present Student List:

The system queries the attendance records and associates them with the corresponding student records. The resulting student names are displayed in the dashboard so that the administrator or authorized user can immediately see who has been marked present.

CHAPTER 8: TESTING

8.1 API Testing

The backend APIs of the Face Attendance Management System were tested to verify that requests are correctly received, validated, processed, and returned to the frontend. Authentication, student-related operations, attendance recording, and attendance retrieval operations were checked during development.

API testing helps verify that the backend behaves correctly independently of the visual frontend. Invalid requests were also considered to ensure that incorrect or incomplete information does not result in invalid database records.

8.2 Integration Testing

Integration testing was performed to verify the interaction between the frontend, backend, face-recognition functionality, and MySQL database.

The complete attendance workflow was tested as an integrated process:

Login → Access Attendance Page → Capture Face → Identify Student → Record Attendance → Retrieve Attendance Data → Display Student Name and Attendance Statistics

This testing ensures that the different modules communicate correctly and that information generated during one stage of the process is correctly passed to the next stage.

The database was also checked after attendance operations to confirm that successfully recognized students were correctly associated with their attendance records.

8.3 Tools Used for Testing

- **Postman** — API testing and verification of backend requests and responses
- **MySQL Workbench** — direct verification of student and attendance records in the database
- **Browser testing** — verification of the frontend interface and user interaction
- **Camera/Webcam testing** — verification of face capture and student identification
- **Manual testing** — verification of login, attendance, dashboard statistics, and student-name display

8.4 Test Cases and Results

A representative set of test cases for the Face Attendance Management System is shown below.

#	Test Case	Expected Result	Actual Result
1	Login with valid credentials	User is successfully authenticated and redirected to the application	Pass

#	Test Case	Expected Result	Actual Result
2	Login with incorrect password	Login is rejected and an error message is displayed	Pass
3	Access protected page without authentication	Unauthorized access is rejected	Pass
4	Register a new student	Student information is stored successfully in the database	Pass
5	Capture a registered student's face	Face input is captured and processed successfully	Pass
6	Recognize a registered student	Correct student is identified and the student's name is displayed	Pass
7	Record attendance for a recognized student	Attendance record is successfully stored in the database	Pass
8	Recognize the same student again during the same attendance session	Duplicate attendance is prevented according to the system's attendance rule	Pass
9	Display total student count	Dashboard displays the correct number of registered students	Pass
10	Display present student count	Dashboard displays the correct number of students marked present	Pass
11	Display names of present students	Dashboard displays the names associated with present attendance records	Pass
12	Search/retrieve attendance information	Correct attendance information is returned from the database	Pass
13	Submit incomplete login information	Request is rejected with an appropriate validation message	Pass
14	Database unavailable during an operation	Application handles the database error without creating an invalid attendance record	Pass

The testing process helped verify the complete flow of the application, from authentication and face capture to attendance storage and dashboard presentation.

CHAPTER 9: DEPLOYMENT

9.1 Steps to Deploy

The Face Attendance Management System can be deployed locally for development and demonstration using the following general steps:

- Install Node.js and npm on the development computer.
- Install and configure MySQL.
- Create the project database using the provided database/schema or SQL commands.
- Configure the database connection details required by the backend.
- Install the backend dependencies using `npm install`.
- Start the backend server using the project's configured start or development command.
- Start the frontend application using the configured frontend command.
- Open the application in a modern web browser.
- Log in using a registered account.
- Allow camera access when requested by the browser.
- Use the attendance interface to capture and identify registered students.

9.2 Environment Configuration

The application's configuration contains the information required for connecting the backend to the MySQL database and running the server. Database credentials and other environment-specific values should be maintained separately from the main source code so that sensitive information is not unnecessarily exposed.

Typical configuration values include the database host, database port, database username, database password, database name, and server port.

An environment configuration file can be used during local development, while placeholder configuration values can be provided separately for other developers without exposing actual database credentials.

9.3 Hosting of Frontend and Backend

For development and demonstration purposes, the frontend, backend, and MySQL database can run locally on the same computer. The frontend is accessed through a web browser, while the backend processes API requests and communicates with MySQL.

The camera connected to the computer provides the input required for face-based attendance. This local architecture is suitable for demonstrating the complete attendance workflow without requiring cloud deployment.

For future deployment, the frontend and backend can be hosted on appropriate web servers or cloud platforms, while the MySQL database can be deployed using a managed or hosted database service.

CHAPTER 10: CHALLENGES & LIMITATIONS

10.1 Issues Faced During Development

- **Face recognition accuracy:** Face-based identification can be affected by lighting conditions, camera quality, face angle, distance from the camera, and changes in a student's appearance.
- **Camera access:** The browser requires permission to access the camera. If camera permission is denied or the camera is unavailable, the face attendance process cannot operate normally.
- **Database connectivity:** The backend depends on a working connection to the MySQL database. Database configuration or server availability issues can prevent student and attendance information from being retrieved or stored.
- **Duplicate attendance:** The system needs to ensure that repeatedly recognizing the same student does not unnecessarily create multiple attendance records for the same attendance session.
- **Real-time dashboard updates:** The dashboard must retrieve the latest attendance information so that the total present count and list of present students remain consistent with the database.
- **User authentication:** Access to student and attendance information must be restricted to authorized users.

10.2 Solutions Applied

Face-recognition issues were addressed by using registered student information during the identification process and providing a controlled camera-based attendance workflow.

Camera-related issues were handled by requesting browser camera permissions and providing appropriate feedback when the camera cannot be accessed.

Database-related issues were handled through backend validation and database connection handling so that database operations are performed through the server rather than directly from the frontend.

To maintain attendance consistency, the attendance process checks the student's existing attendance information before creating a new attendance entry according to the application's attendance rules.

The dashboard retrieves attendance information from the database rather than maintaining a separate manually entered list. This ensures that the displayed student count and present-student names are based on stored attendance records.

10.3 Current Limitations or Known Bugs

- Face recognition may not identify a student correctly under poor lighting or when the face is partially obscured.
- The system depends on the availability and quality of the computer's camera.
- The current implementation is primarily designed for local development and demonstration.

- A large-scale deployment involving many simultaneous users may require additional optimization and infrastructure.
- Advanced anti-spoofing or liveness detection may not be available in the current implementation.
- Automatic notifications such as email or SMS are not part of the current attendance workflow.

CHAPTER 11: FUTURE ENHANCEMENTS

11.1 Planned Features

- **Improved face-recognition accuracy** using more advanced recognition techniques and improved training or registration procedures.
- **Liveness detection** to distinguish a real person from a photograph or displayed image during attendance.
- **Multiple attendance sessions** allowing different classes, subjects, or periods to maintain separate attendance records.
- **Attendance reports** that allow administrators to generate daily, weekly, monthly, and semester-wise reports.
- **Attendance percentage calculation** for individual students.
- **Search and filtering** of attendance records by student, date, class, or subject.
- **Email or SMS notifications** for attendance-related alerts.
- **Student profile management** allowing authorized administrators to update student information.
- **Export functionality** for downloading attendance records in formats such as CSV or PDF.
- **Role-based dashboards** with different views for students, faculty, and administrators.

11.2 Possible Integrations or Optimizations

- Integrating an advanced face-recognition model to improve identification accuracy under different lighting and camera conditions.
- Adding a liveness-detection mechanism to reduce attendance fraud using photographs or recorded videos.
- Optimizing database queries and indexes for faster retrieval when the number of students and attendance records increases.
- Adding caching for frequently requested dashboard statistics.
- Deploying the application using Docker to provide a consistent environment across different computers.
- Hosting the frontend, backend, and database on cloud infrastructure for access from multiple classrooms or locations.
- Adding automated unit and integration tests to reduce the possibility of regressions during future development.
- Integrating the system with an existing college student information system to avoid duplicate student registration.

CHAPTER 12: CONCLUSION

12.1 Summary of the Project

The Face Attendance Management System was developed to automate the traditional attendance process by using face-based student identification and a database-driven web application. Instead of requiring faculty or administrators to manually call student names or maintain paper-based attendance sheets, the system captures a student's face, identifies the registered student, and records the attendance electronically.

The system combines user authentication, student information management, face-based identification, attendance recording, and dashboard-based monitoring into a single application.

12.2 What Was Achieved

The completed system provides an automated approach to recording student attendance. Users can authenticate themselves and access the attendance functionality, while registered students can be identified through the camera-based face-recognition process.

Once a student is successfully recognized, the system records the attendance information in the database and displays the student's name. The dashboard provides an overview of the attendance status by displaying the total number of students, the number of students present, and the names of students who have been marked present.

The database provides centralized storage for student and attendance information, making it easier to retrieve and manage attendance records compared with traditional manual methods.

12.3 Skills Learned

Developing this project provided practical experience in relational database design, backend development, frontend development, API communication, user authentication, and database management.

The project also provided experience in integrating face-recognition functionality with a web-based application and connecting the recognition process with database-driven attendance records.

Additional skills gained include designing user interfaces, handling camera input, validating user requests, managing database operations, testing application workflows, and troubleshooting communication between the frontend, backend, and database.

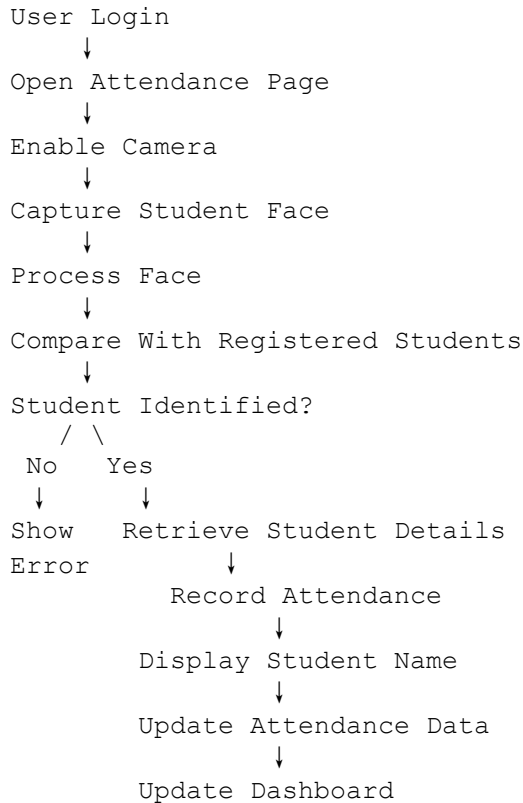
CHAPTER 13: REFERENCES

- Node.js — Official Documentation
- MySQL — Official Documentation and MySQL Reference Manual
- Express.js — Official Documentation
- JavaScript — MDN Web Docs
- HTML and CSS — MDN Web Docs
- Git — Official Documentation
- GitHub — Official Documentation
- Postman — Official Documentation and Learning Center
- MySQL Workbench — MySQL Documentation
- Documentation of the face-recognition library/model used in the project
- Documentation of the frontend framework/library used in the project

CHAPTER 14: APPENDICES

Appendix A: Sample Attendance Workflow

The following workflow represents the main attendance process implemented by the Face Attendance Management System:



Appendix B: Installation and Setup Instructions

- Install Node.js and npm.
- Install MySQL and create the required database.
- Import or execute the project's database SQL file.
- Configure the database connection in the backend configuration.
- Open the backend project directory.
- Install the required dependencies using `npm install`.
- Start the backend server using the configured command.
- Open the frontend project directory.
- Install the frontend dependencies using `npm install`.
- Start the frontend application.
- Open the application in a supported web browser.

- Allow camera access when prompted.
- Register or use an existing authorized account.
- Register the required student information and face data.
- Open the attendance interface and perform face-based attendance.

Appendix C: Login and User Information

The application requires authenticated access to protected functionality.

The following information should be documented according to the actual accounts created for the project:

Role	Account	Access
Administrator	Project-configured admin account	Student management, attendance monitoring, dashboard
Authorized User	Project-configured user account	Attendance-related functionality
Student	Registered student account, if applicable	Student-specific functionality

Note: Actual usernames, email addresses, and passwords should be entered here based on the accounts configured in the project's database. Passwords should not be included in the final report if the report will be publicly shared.

Appendix D: API Reference Summary

The backend API provides communication between the frontend and the server-side application.

The major API operations include:

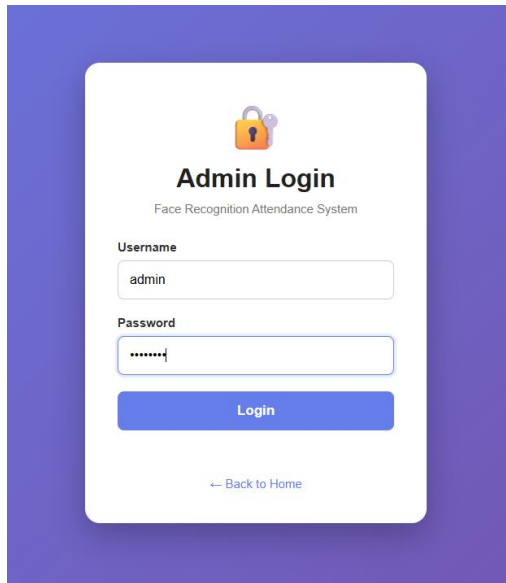
Resource	Operations
Authentication	Register, Login, Verify authenticated user
Students	Add, retrieve, update, and manage student information
Face Recognition	Process face input and identify a registered student
Attendance	Create, retrieve, and manage attendance records
Dashboard	Retrieve total students, present count, and present student names

The exact endpoint paths and request/response formats should be included according to the final backend implementation.

Appendix E: Screenshots

The following screenshots can be added to the final project report from the running Face Attendance Management System:

1. Login Page

A screenshot of the Admin Login page. The page has a purple gradient background. In the center is a white rounded rectangle containing a lock icon, the title "Admin Login", the subtitle "Face Recognition Attendance System", a "Username" field with "admin" entered, a "Password" field with masked characters, a blue "Login" button, and a "← Back to Home" link.


Admin Login
Face Recognition Attendance System

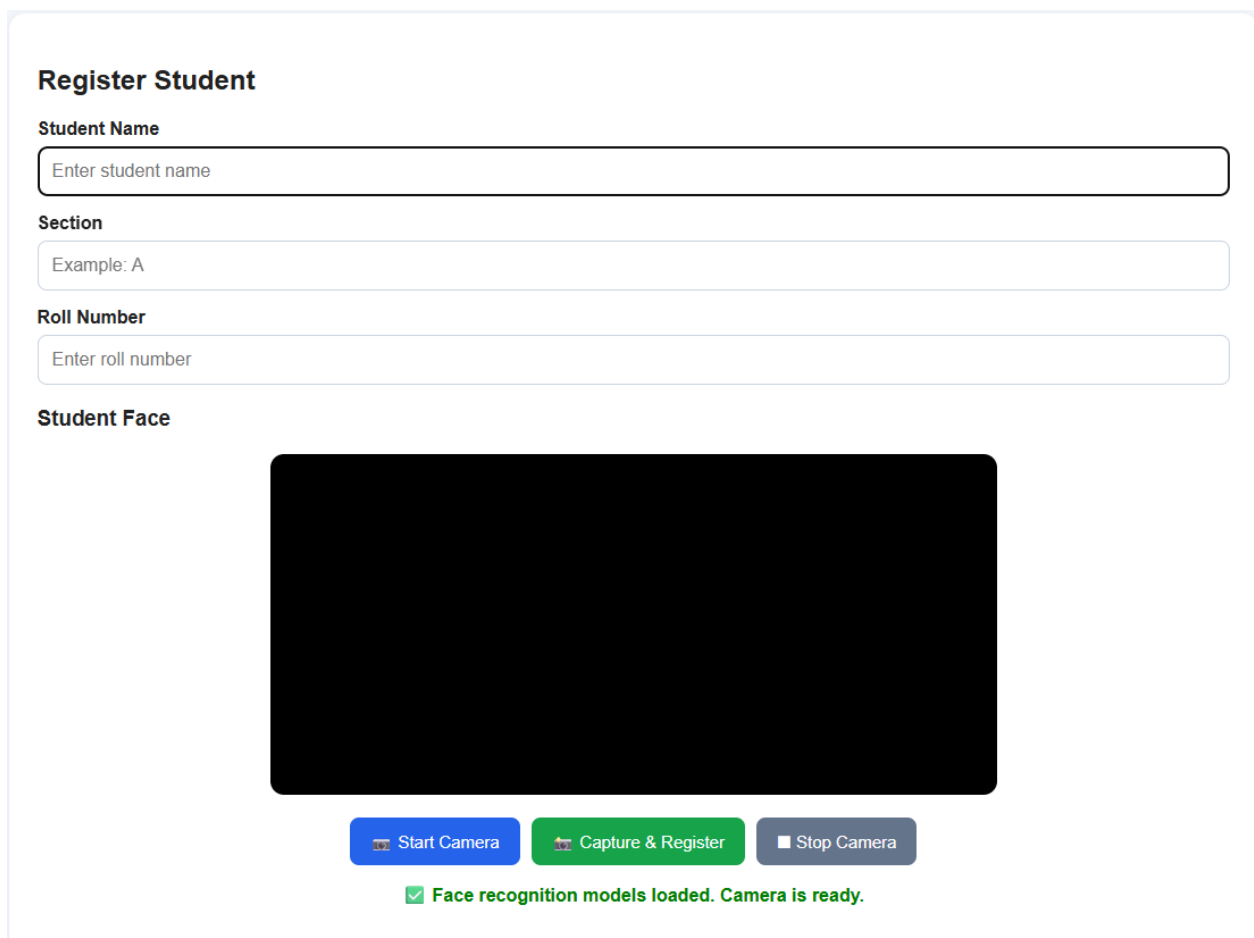
Username

Password

[Login](#)

[← Back to Home](#)

2. Registration Page

A screenshot of the Register Student page. The page has a light gray background. It contains a title "Register Student", three input fields for "Student Name", "Section", and "Roll Number", a large black rectangle for the "Student Face", and three buttons: "Start Camera", "Capture & Register", and "Stop Camera". A green status message is at the bottom.

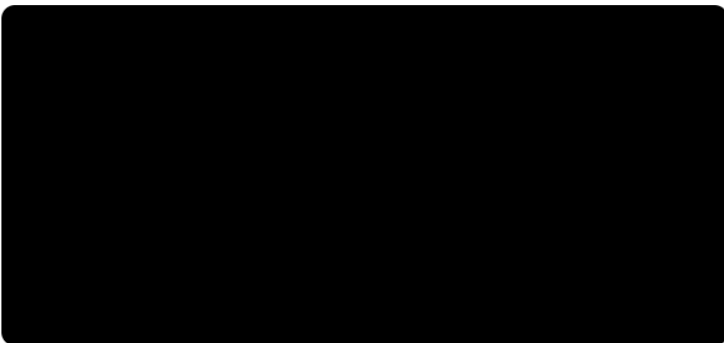
Register Student

Student Name

Section

Roll Number

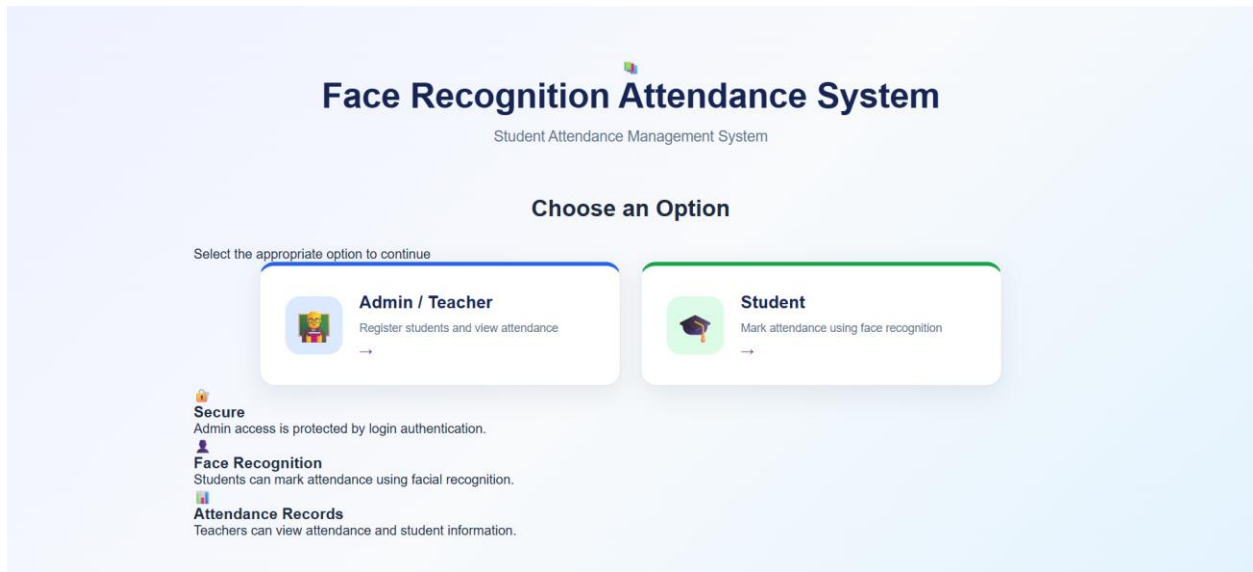
Student Face



[Start Camera](#) [Capture & Register](#) [Stop Camera](#)

✔ Face recognition models loaded. Camera is ready.

3. Dashboard



4. Student Registration Page

Register Student

Student Name

Section

Roll Number

5. Face Attendance Page

Mark Your Attendance

Look directly at the camera and make sure your face is clearly visible.

Start Camera

Mark Attendance

Stop Camera

Models loaded. Camera is ready.

[← Back to Home](#)

6. Successful Face Recognition

- Camera started. Position one face inside the camera.
- Student registered successfully!

7. Attendance Records

Total Students	Present Today	Absent Today
0	0	0

8. Present Students List

Students Present Today

#	Name	Section	Roll Number	Time
1	Shiva	10	2520090092	01:52:11 pm

Attendance Records

Name	Section	Roll Number	Date	Time
Shiva	10	2520090092	2026-10-02	01:52:11 pm

9. Database Tables

Attendance Records

Name	Section	Roll Number	Date	Time
Shiva	10	2520090092	2026-10-02	01:52:11 pm

10. API Testing

Attendance Marked

Name: Shiva

Section: 10